

SHRUTI VERMA

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AEE, IRCA, BEE Certified Energy Manager, Lead Auditor as well as a certified Boiler Operation Engineer (BOE) bestowed by the Govt. of Chhattisgarh, 12+ years of experience in Asset Performance Management (APM), Energy Audit, Energy Efficiency Management, O&M Consultancy Services, design reviews of Electrical, Mechanical, BTG and BOP (Balance of Plant) packages.

Professional Experience

Present Employer: Black & Veatch, Pune - India (June 2018 till present)

- I am presently associated as **Lead Asset Management Consultant (sub - leader O&M Consulting)** in the AMS Section of the Power Generation Services business.
- My main responsibilities include O&M and Reliability (like Life extension, RLA, O&M estimates, FMEA and risk assessments, Conceptual study, design review and Strategy, Technical due diligence, Electrical safety, Owners Engineer and MFS preparation, Benchmarking etc.), Proposal preparation, etc.
- Prior to this, I have led a team at the Monitoring & Diagnostic Center in Pune focusing on Remote Asset Management, Fault Detection and Diagnostics and executing expert layer of services for various types of assets spanning across Electrical systems, Thermal and Combined Cycle (Supercritical and Sub critical), Solar, Wind, Data center, Gas and Microgrids facilities of around 65 GW cumulative capacity.

Previous Employers:

1. Bharti Infratel Limited (From September 2017 to June 2018)
 - **Responsibilities:** Energy Audits, Electrical O&M (Transformers, BESS, EB, Switchgear, Substations, Grounding and Lighting etc.), Energy optimization of Solar cells, Lithium Ion-Lead acid battery, DG, SMPS and transformer, Energy Management, Data analytics, Designing and Process Improvements, BESS, etc.
2. KSK Mahanadi Power Company Limited (February 2014 to September 2017)
 - **Responsibilities:** Efficiency & Energy Management of 3600 MW coal-based plant, Desk and Electrical operations (Switchyard, ESP and Switchgear, BTG O&M, Maintenance planning and designing and Commissioning
3. Adhunik Power and Natural Resources Limited (August 2010 to February 2014)
 - **Responsibilities:** Project Management, Commissioning of Electrical and BTG area, SAP PM implementation and MM hands on experience, Auto cad designing of Key SLD'S and load list, Operation and Maintenance, Design review etc.

Accolades at Workplace

1. Best Employee of the Month in Black & Veatch Pune in 2021
2. Spot Award in Black & Veatch in 2020
3. O&M Engineer award in KSK.
4. Awarded as Best GET in year 2011.

Certificates

- IRCA certified 50001 External Energy Auditor
- Accredited as Energy Auditor by AEE
- Accredited as Energy Manager by BEE
- Diploma in Electrical Audit and Safety
- Boiler Operation Engineer Certified by CG Boiler Inspectorate.
- Certified Young Manager from CII
- Completed One-year program in Data Science from ETL Hive
- ISO internal auditor in year 2014 from Bureau Veritas.
- Pursuing PGP in AI-ML from The University of Texas at Austin
- Undergoing Project Management Training in PERT & CPM.

Educational Credential

Course	Institution	% Of Marks Secured	Year of Passing
M.TECH (Power System)	Dr. C.V Raman University, Bilaspur	8.5-CPI	2016
Post Graduate Diploma in Thermal Power Plant Engineering	National Power Training Institute (NPTI), Durgapur	82.2%	2010
B.E. (Electrical Engineering)	Government Engineering College, Bilaspur	8.3-CPI	2009

Personal Information

Date of Birth : Sept. 8th, 1987
Nationality : Indian
Gender : Female
Marital Status : Married
Language Proficiency : English, Hindi.
Personal Number 1 : 9893583088

Declaration

I hereby vouch for the authenticity of the above information to be true and correct to the best of my knowledge and belief, till date.

Date: 4th July-2022

Place: Pune

Address: I-1002, Park Royale CHS, Rahatani, Pune, India 411017

Shruti Verma

BLACK & VEATCH PVT LIMITED

- Currently responsible for dealing with clients across the globe and providing consulting services for Operation and Maintenance, Substation RLA, Owners Engineering, providing O&M Estimates and Strategy, sharing O&M best practices and Risk Matrix, Interlock Protection and Controls Review, Logic preparation and modification, Plant modernization of BTG and Combined Cycle, R&M, Monitoring and Maintenance of Distributed Energy asset, Control System, feasibility study of Substation, BESS, Solar PV, DG's, Wind etc.
- Electrical Design Engineering (conceptual and secondary), Bid preparation and Technical specification, evaluation and vendor drawings and One-line diagrams, three-line diagram, Relay & metering diagrams, AC / DC schematics and wiring drawings, CT/PT sizing calculations, SAS Architecture and layouts
- Failure, Losses and Performance Modelling analysis of Solar, Wind, Microgrid, Control System (GT and other controls) etc.
- Cable and transformer loss calculation, Arc flash, Solar Substation and TL loss, Battery sizing and DC sizing calculation, Control system feasibility and other evaluation (controls, etc. for different utilities.
- Coordinating with external clients for project prospects and scope clarification, propose a solution and ensuring that projects are executed within the committed schedule and cost to meet the requirements.
- Previously have led a team responsible for remote monitoring and performance analysis of Boiler, Turbine Inverter, HVAC system, Solar (PV, Solar field, BESS etc.), Microgrid, cooling load estimation, Heat exchangers, Compressors Domestic/Commercial Refrigerators, Air-Conditioning Units such as ductable Split Units, Air Handling Units, Chiller systems and Solar assets (Polycrystalline, Parabolic trough HTF, CIGS etc.), Wastewater facility, Paper and pulp, 3 DG Gensets (reciprocating engines), Combined cycle (GT) and crude facility.
- Was also responsible for Proposal preparation of various APM & O&M prospects. Offer evaluation, interaction with supplier, Bid Evaluation and planning for APM implementation in different sites
- Carried out control system requirements definition (Mark VI, Vie, Max DNA etc.) and its condition assessments, assistance on Communications protocols like -OPC DA, TCP -IP, FTP etc., AWS hosting evaluation etc.

BHARTI INFRATEL LIMITED

- Single point of contact for virtually all aspects of energy management prerequisite in Raipur zone for carrying out energy audits /efficiency improvement activities by analyzing technical and financial feasibilities of suggestions made and following up for implementation of these recommendations.
- Conducting energy analysis, electrical maintenance & benchmarking the consumption pattern of all energy equipment's.
- Auditing the performance of the DG sets installed at sites with manufacturer specification, tracking the records of fuel consumption at site, and calculating the fuel efficiency of the DG sets as a derivable parameter.
- O&M and electrical maintenance of transformers, BESS, Solar PV units, DG, SMPS, Load Commutated Inverters, Control Systems, HVAC, EB etc.
- Coordinating with external agencies for technical queries and scope clarification, propose a solution and ensuring that projects are executed within the committed schedule and cost to meet the requirements.
- Working on conversion of existing DG sites into green sites by proper analysis of energy consumption pattern, and available energy from electricity board
- Ensuring proper functioning and calibration of instrumentation required to assess level of energy consumption directly or indirectly.
- Preparing energy efficiency reports, energy supplies quality analysis report for review meetings.
- Performance analysis of HVAC system, Solar and BESS asset, cooling load estimation, Heat

exchangers, Compressors Domestic/Commercial Refrigerators, Air-Conditioning Units such as Ductable Split Units, Air Handling Units and Chiller systems.

- Proposing innovative systems & procedures is to achieve energy savings without affecting service quality.
- Daily Monitoring & Preparation of daily, weekly and monthly MIS report.
- Performing Technical investigation & Root Cause Analysis.
- Remote monitoring of sites for better control over energy spending.

KSK MAHANADI POWER COMPANCY LIMITED

- Commissioning, Operation & Maintenance of BTG – BOP packages
 - BTG and Electrical (Switchgear, Generator, Controls and 400 KV Switchyard) Operation and maintenance, Outage Assessment and field operational activities
 - Was responsible for managing commissioning activities and other maintenance jobs of the Boilers, Turbine and other Balance of Plant equipment's during plant shut down.
 - Responsible for Interlock and Protection review, logic modifications, Ovation DCS and controls
 - Was responsible for the Performance Guarantee Test of Unit-3 (Electrical, Boiler, Turbine & APC)
 - Was involved in commissioning & Reliability Run Test of Unit-4 including boiler light up, initiation of turbine barring gear, Turbine and Boiler Commissioning, Outage Assessments etc.
- Power plant process auditing and energy efficiency management
 - Analyzed the various processes at site, identified possible leakages in those processes and suggested preventive measures to be implemented for optimizing these processes.
 - Carried out efficiency calculation of various key components such as Boiler, Generators, Transformers, Switch gears, Turbine, Heaters, APH, CT fans, Pumps etc and identified possible reasons for deviation from plant guaranteed parameter. Prepared and presented these observations and suggestions to the top management for their perusal.
 - Prepared and presented various reports (such as Generation loss monitoring, trip report, startup report, exception report, ORM report) to the top management including the MD of the company and assisted them in proper decision making to ensure optimized economic performance of the plant.

ADHUNIK POWER AND NATURAL RESOURCES LTD.

- Commissioning and Operation of 2 x 270 MW Turbine – Generator and its auxiliaries
 - Worked as Commissioning Engineer of 2X 270 MW BHEL thermal power plant (MAX DNA DCS) basically looking after Transformer, Generator and BTG package. Also, actively involved as the lead engineer and coordinate with Bhel, ABB, Siemens and its subcontractors. Visited for FAT test and design finalization to shops.
 - Was responsible for checking of permissive interlocks & protections for various equipment from DCS and key major commissioning activities like Steam Blowing, Lube oil flushing, commissioning of Generator, Switchgear, Transformer (Testing's, Erection and commissioning tests. Interlock, Protection and logic modifications preparation in Max DNA etc.
- Supply Chain Management, Project Monitoring and Control
 - Co-ordinated with site engineers and the vendors, controlling the contractors and supervising their work to ensure on time completion of project as per the contract guidelines.
 - Worked as SAP-PM (Preventive maintenance team core member) and PM implementation team for SAP material management module application
 - Was responsible for budgeting and scheduling of various activities at site by supervising the supply chain inflow and planning for optimized materials management.
 - Was responsible for verifying the Supplier's Bills as well as Service Bills and maintaining the record for future tracing.

- Design Engineering
 - Was involved in the evaluation of technical bids from various vendors to ensure selection of the most suitable contractor's for designing, planning, delivery, erection and commissioning of various especially BTG (TG and Electrical) packages
 - Electrical Design Engineering (conceptual and secondary), Bid preparation, evaluation and vendor drawings and One-line diagrams, three-line diagram, Relay & metering diagrams, AC / DC schematics and wiring drawings, CT/PT sizing calculations, SAS Architecture and layouts
 - Carried out inspection and assessment of material at the vendor's facilities including technical discussions, quality inspections, drawing & technical data finalization, etc. Was responsible for management of all the necessary documents such as EIFS and Drawings & necessary engineering modifications required during setting up of the power project.
 - Assessed the technical/ engineering problems in site and solved these problems with the help of drawing study.
 - Followed up with vendors on the delivery of materials required at site for smooth execution of the project

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